

Stiff-switch: Finite Stiffness Control Method Using a Thermo-Mechanical Structure and a Low-voltage Pinpoint Heater

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ABSTRACT

We propose Stiff-switch, thermo-mechanical stiffness control method. Phase change material (PCM) is often used to control stiffness by heat. However, high-voltage requirements and slow heating speed still limit the practical application. We present new thermo-mechanical structures that can efficiently control the stiffness by heating the minimum portion of the structure where stress is concentrated. We employed phase change materials in an H-beam structure and mechanical linkage joint to control stiffness against linear stress and torque, respectively. Additionally, we investigated carbon-based electrically heating paint as a novel joule heating method to achieve fast temperature control speed while requiring low voltage. We demonstrate the proof-of-concept by deploying a PLA filament and hot melt as phase change materials and experimentally show the wide-range stiffness control capability and robust shape-recovery function from large deformation. Finally, we discuss the possible future application scenario for airless tires, tactile tablet pens, robot hands, and finger exoskeletons.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

elasticity, stiffness switch, phase change material, shape memory, paper interface, heating paint, 3D printing, metamaterial, structural mechanics

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1 INTRODUCTION AND RELATED WORKS

With the prevailing 3D printing technologies and smart materials, the design of an object's stiffness becomes more flexible and controllable in personal fabrication, product design, and mechanical engineering. For instance, researchers explored a variety of soft

3D printing techniques to fabricate a 3D object with the desired flexibility and softness perception by tuning the infill density and pattern within the volume of a target 3D model [8, 21, 33].

The microstructural design of mechanical metamaterial [2, 11, 23, 27, 30] enables the finite stiffness control of the deformable 3D objects by utilizing a single rigid material and facilitates the fabrication of functional footwear [3, 4, 35] and automobile tire [13]. Yang et al. propose the concept of reconfigurable elastic metamaterial composed of damper unit cells and spring unit cells [35]. By arranging two different elastic cells in the insole, the end-user can customize the heterogeneous elasticity of the footwear optimized for the style of exercise (e.g., jump, run). The concept of reconfigurable elasticity suggests a plethora of opportunities to break the limit of static materials and create an unusual physical experience beyond the rule of Classical Mechanics [25, 35].

However, dynamical stiffness control with 3D-printed objects has still remained the key challenge in the field of product design and human-computer interaction. Researchers explored variable stiffness with 3D-printed objects by deploying thermal-responsive phase change material (PCM)[16, 19, 22, 29, 31]. Low-temperature thermoplastics, such as Poly-Caprolactone (PCL) and Poly-Lactic Acid (PLA), are widely used phase change materials in Fused Deposition Modeling (FDM) 3D printers and 3D pens. Researchers explored the thermal stiffness control method to enable shape manipulation of 3D printed models by utilizing PLA and PCL [19, 29] and investigated the dynamical stiffness control method by embedding heating elements in 3D printed structures [16, 22, 31]. As other low-temperature phase change materials, wax[7] and liquid metal alloy[32] are also used to fabricate variable stiffness soft robots.

Although thermal stiffness control methods with phase change materials have been widely explored, the requirement of high-voltage resistive heating elements and a long heating cycle to melt phase change materials limit the application to slab deflection control and 3D model surface manipulation [16, 22]. The robustness of interactivity and sustainability of stiffness control has not been achieved with the 3D printed model due to the thermo-plastic deformation, which causes the loss of elasticity necessary for the recovery of the original shape after deformation.

To address the constraints mentioned above, we propose the Stiff-switch, new thermo-mechanical structures that can efficiently control the stiffness by heating the minimum portion of the structure where stress is concentrated. The Stiff-switch also allows for repetitive control of stiffness for large deformation. We introduce the fabrication method of the high-speed and low-voltage heater by deploying carbon-based electrically heating paint as a customizable joule heating element. We explore thermo-mechanical structures that require a minimum area to heat phase change material but still

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reduce structural strength significantly. In particular, we propose two kinds of Stiff-switch: H-beam stiffness switch and linkage-joint stiffness switch. H-beam stiffness switch utilizes a thin phase change material to control buckling strength based on temperature and achieve rapid stiffness change with large deformation. Linkage joint switch employs a phase change material within a linkage joint to regulate torque and rotation of mechanical linkage. By controlling the friction in a pivot joint with the temperature-dependent adhesiveness of the phase change material, the device can achieve stiffness control in the linkage mechanism. We introduce the fabrication method and demonstrate the proof-of-concept of Stiff-switch with 3D FDM printers, using PLA filament and hot melt adhesive as phase change materials. Finally, we will discuss potential future applications of the Stiff-switch, including adaptive airless tires, tactile tablet pens, robot hands, and finger exoskeletons.

2 STIFFNESS SWITCH

The motivation of the stiffness switch is three-fold: (1) utilize a minimum amount of phase change material to cause significant stiffness change with low joule heating voltage (2) explore simple stiffness control mechanism based on stress distribution (3) control stiffness reversibly and robustly against the series of user interaction. To fulfill these requirements, we introduce thermo-mechanical structures that effectively use phase change materials based on stress distribution. Furthermore, we explore a novel low-voltage joule heating method by deploying carbon-based electric heating paint.

2.1 Thermo Mechanical Structure

We explore two kinds of thermo-mechanical stiffness switches: H-beam stiffness switch (Figure 1) and linkage joint stiffness switch (Figure 2).

2.1.1 H-beam Stiffness Switch. H-beam is a standard structural shape of steel used in construction and building [6]. The H-beam is a structurally efficient design that effectively resists yielding, bending, and vibration while using a minimal volume of materials. As a result, it is often employed in the construction of large buildings to reduce material weight and cost. As (Figure 1 (a)) shows, the stress distribution of composite materials including the H-beam structure is concentrated on the central pillar, which thereby defines the structural stiffness based on the buckling strength. Hence, we can deploy H-beam structures to minimize the amount of phase change material needed and efficiently regulate stiffness by solely heating the phase change material within the pillar instead of heating the entire body. Figure 1(b)-(d) shows the working principle of H-beam stiffness switch. The H-beam stiffness switch is composed of a pillar made of phase change material and a joule heating element. It is combined with a hyper-elastic frame and silicone to provide shape recovery capability from large deformations. In the hard mode (Figure 1 (b)) when the joule heating element is inactive at no voltage, phase change material is solid and resists large stress, producing high structural stiffness. In the soft mode (Figure 1 (c)), a voltage is applied to heat phase change material to melt. H-beam loses buckling strength as the pillar is unable to support the large load stress and decreases the structural stiffness significantly. By infilling the hyper-elastic soft silicon around the pillar, the H-beam structure can recover the original shape after the release of stress

while phase change material is liquified at high temperature, and regains the high stiffness in the hard mode as the temperature decreases (Figure 1 (d)).

2.1.2 Linkage Joint Switch. While the H-beam is an effective thermo-mechanical structure for controlling stiffness against linear stress, it is not applicable to controlling stiffness for torque. We introduce the linkage joint switch as another thermomechanical structure that can control torque stiffness to regulate the pivot of mechanical linkages (Figure 2). If a torque is applied to linkage frames, the stress is concentrated at the pivot joint of two frames. By applying hot melt glue or wax to the pivot joint, we can control the torque stiffness by heating only the joint space. In the hard mode when phase change material is a solid state at room temperature (Figure 2 (a)), we need high torque to rotate one of the frames against the friction of phase change material in a joint. In the soft mode, a voltage is applied to the joint heater to melt the phase change material (Figure 2 (b)), and facilitates the frame rotation with a little torque stiffness. We can deploy linkage stiffness switch in multiple pivot joints and control the stiffness and freedom of motion in mechanical linkage mechanisms with a minimum heating power.

2.2 Carbon-based Low-voltage Heater

To generate enough heat to melt phase change material at low voltage, we investigated methods to fabricate customizable heaters with carbon-based electrically conductive heating paints, such as G-thermoheld Carbo e-Therm [10, 12] and G-Therm paint by Graphen Star [15, 28]. Since carbon-based conductive paints efficiently generate heat as a form of far infrared radiation energy, it has been used to warm large areas of the building (e.g., underfloor, wall, ceiling) [12, 14] and curved objects (e.g., chairs, pipes, bathtubs) [9]. The unique feature of carbon heating paint is that any painted surface between electrodes can work as a uniform heater with high conductivity around $30 \Omega/sq$ [9, 20]. The uniform heating capability and the low cost of carbon-based paints facilitate custom heater fabrication more simple and approachable than conventional silver conductive ink and electrical wire as those high conductive materials require a serpentine pattern to adjust joule heating resistance based on the heating area [16, 34]. The high conductivity of carbon paints also enables low voltage heating capability with a fast temperature rising speed [9, 20] compared to conductive PLA, which has high resistance ($> k\Omega$) and therefore requires a high voltage in the range of a few hundred volts to generate enough Joule heat to melt the phase change material [22]. Markevicious et al. reported the usability of Carbo e-Therm heating paint to create a large heater ($0.25 \text{ m} \times 0.16 \text{ m}$) for art conservation [20]. To the best of our knowledge, no prior work has been conducted on using heating paint for thermal stiffness control.

3 FABRICATION AND PROOF-OF-CONCEPT

3.1 Carbon-based Custom Heater

Among several available heating paint products, we used flexible polyurethan-based heating paint, Carbo e-Therm PUR 70-350B.01 [9], which has a low resistance rate between 350 and $450 \mu\Omega/m$. We can dilute it with water, up to 25 % by weight, without losing conductivity. Various conductive materials, such as copper foil tape, wire, and

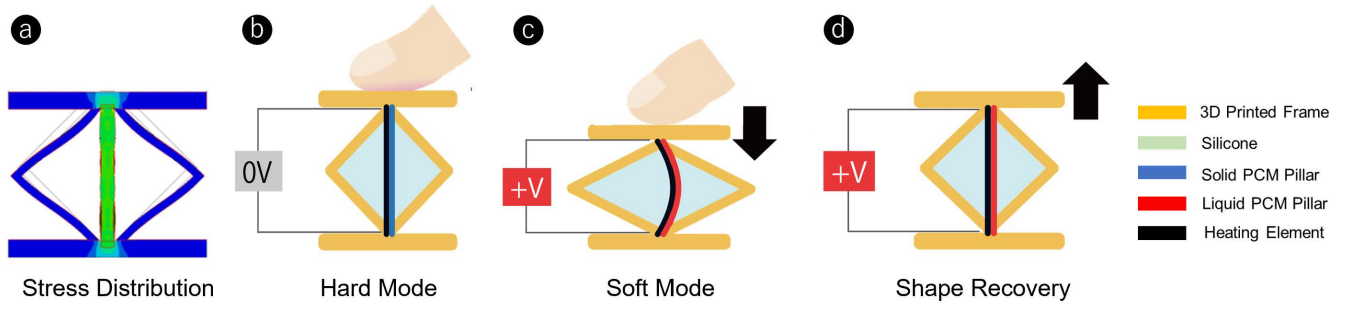


Figure 1: H-beam stiffness switch.

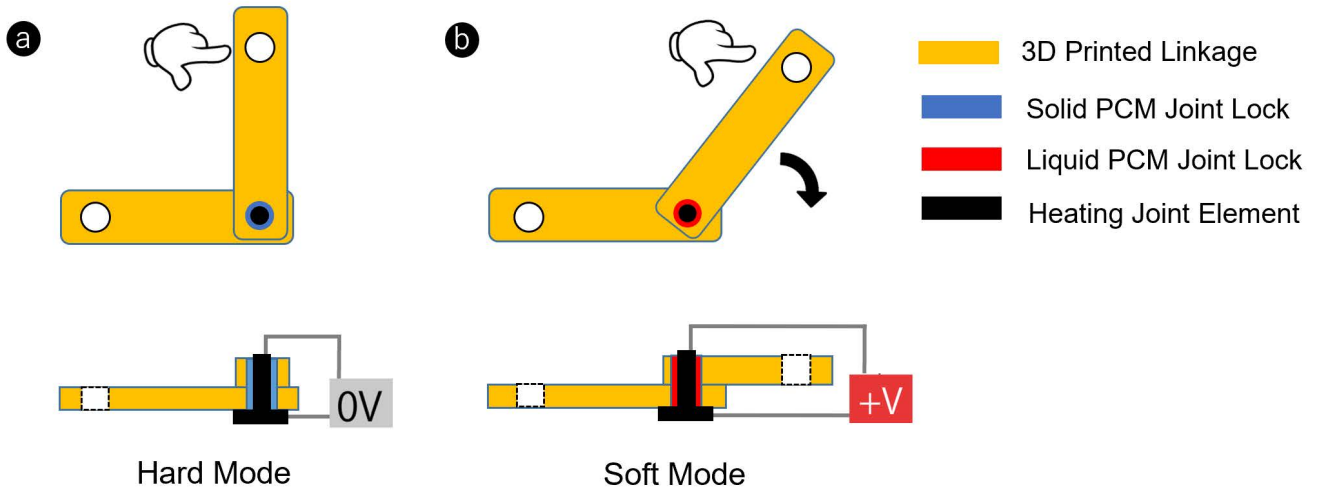


Figure 2: Linkage joint switch.

silver conductive ink, can be used as electrodes. It is recommended to paint both the electrode and heating area to create a uniform heating area with little contact resistance [9, 14, 20]. Like regular paint, we can apply heating paint to diverse materials including glass, FR4, wood, plastic, and paper. In particular, we found paper is the optimal substrate material to paint using a brush and a spray. The flexibility of the paper allows it to conform to the buckling of a pillar in the H-beam stiffness switch. We also found heating paint is compatible with 3D-printed objects and can easily paint on the surface to allow direct heat. However, since the adhesion of 3D-printed surface and heating paint is not strong enough, we found heating paint coating cannot follow the large deformation of the 3D-printed object and cause cracks, which damages the conductive traces and loses high conductivity.

Figure 3 shows the fabrication process of the paper heater. To compare the heating performance with silver-conductive ink heater reported by Groeger et al. [16], we fabricated a heater with the same dimension as their tested size, 20 mm × 10 mm. We cut a standard copy paper with a thickness of 76.8 μm with a craft cutter knife and taped 3 mm width copper foil as an electrode on the short side

of the paper, as Figure 3 (a) shows. The length of white paper space between two electrodes is 2 cm. We then soldered a heat-resistant wrapping wire with a diameter of 32 AWG to the copper electrode as shown in the bottom image of Figure 3 (a).

Since the glossy surface of brand-new copper tape can repel water, we prepared a few milliliters of heating paint in a plastic paint pallet without thinning with water to keep high enough viscosity (3 to 4 Pa·s). We used a 5mm standard oil paint hair brush to paint wires, copper foils, and a copy paper evenly, as shown in Figure 3 (b). We painted one side of the paper attached to wires to keep the flexibility of the paper as the bottom image of Figure 3 (b) shows. While the complete dry time of the heating paint is 50 minutes at room temperature, we found a heat gun can dry the paint in 10 seconds to gain high conductivity. We painted five copy papers to evaluate the variation of conductivity with a brush painting. We measured the thickness with a coating thickness gauge (Wintact WT230) and the resistance with a multimeter (KAIWEETS KM601). The average thickness of dry heating paint in these test samples was $54.6 \pm 9.2 \mu\text{m}$ and the resistance was $18.8 \pm 4.5 \Omega$. In order to safeguard the heating paint from scratches, water, and short

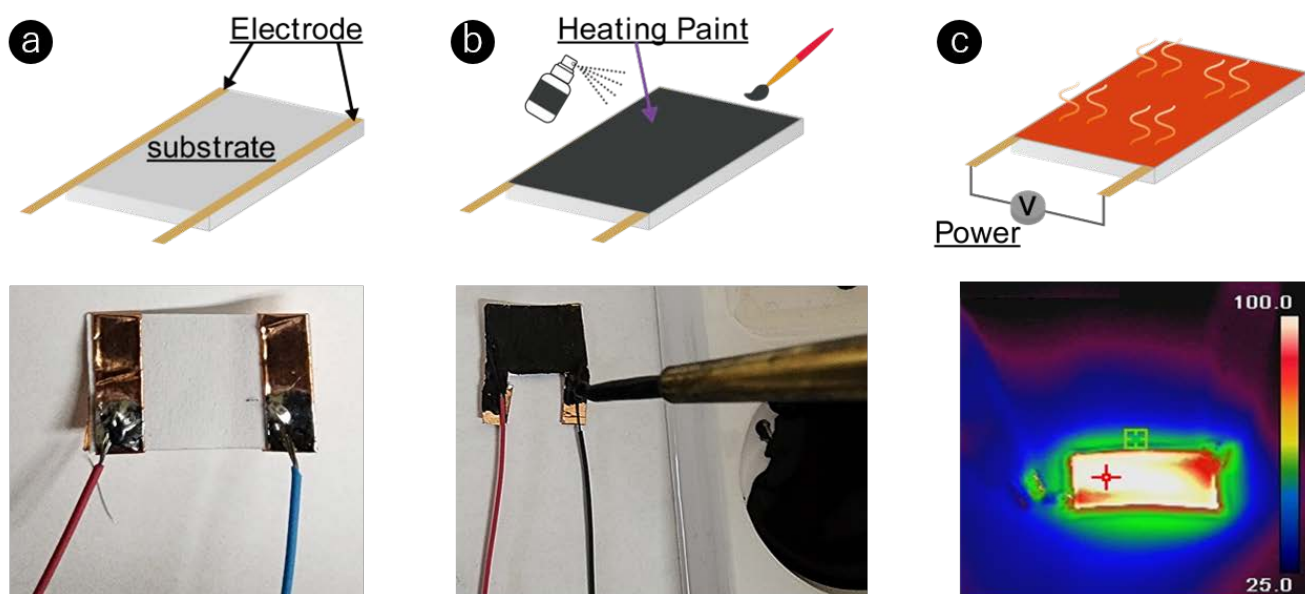


Figure 3: Fabrication process of a heater with electrically heating paint.

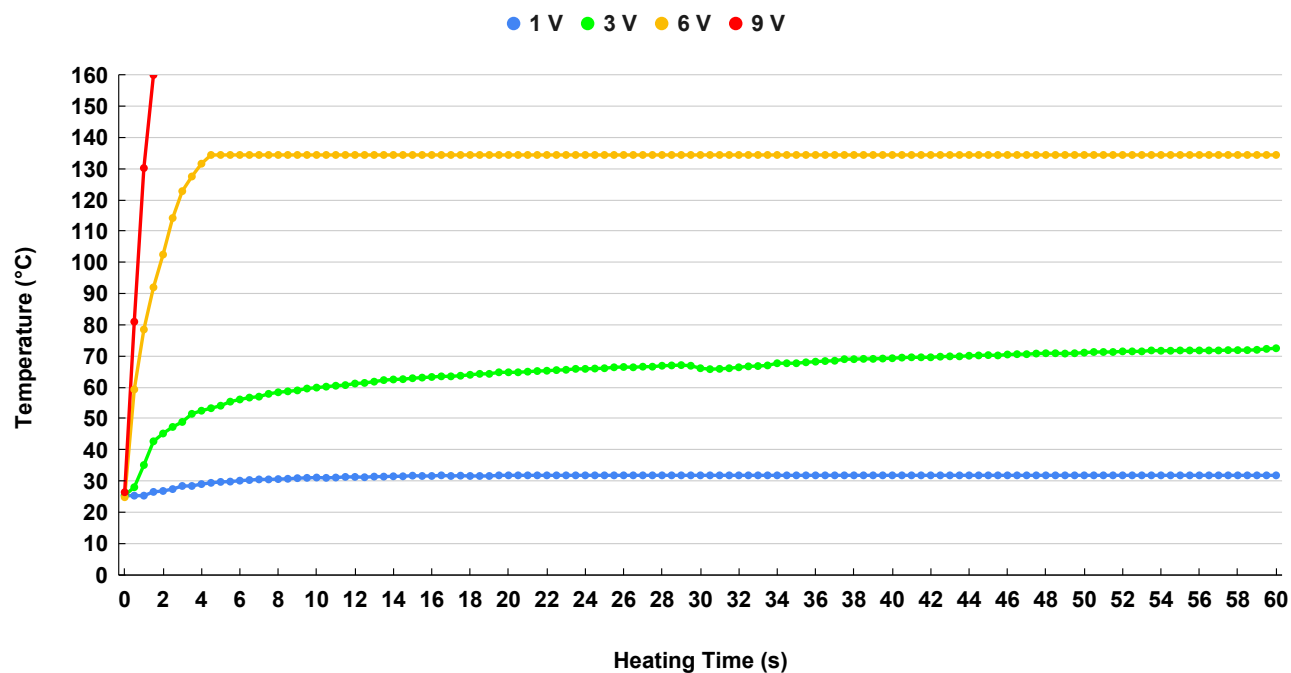


Figure 4: Heating Speed of 2 cm x 1 cm paper heater based on voltages (1 V, 3 V, 6 V, 9 V).

circuits, we applied a heat-resistant, insulating silicon spray to the paper heater, as illustrated in the top image of Figure 3 (b). We applied DC voltage to the electrode through the wire and checked

the uniform heating capability with a thermal camera, as shown in the bottom image of Figure 3 (c).

3.2 Evaluation of Heating Speed

We measured the speed of the rising temperature of the fabricated paper heater. We applied multiple voltages to the electrodes of the paper heater by utilizing a DC-stabilized power supply (Wanptek WPS300) and measured the temperature change with a thermal camera (HIKMICRO B10) at 0.5 second intervals. As Figure 4 shows, the temperature of 2 cm × 1 cm size paper heater increases up to 160 °C within a few seconds at 9 V. The heater gains the maximum temperature of 136 °C at 6 V after 4 seconds, 70.5 °C at 3 V after 60 seconds, and 32.5 °C at 1 V after 20 seconds. Greoeger et al. reported silver-conductive ink heater took 11 seconds to reach a temperature of 60 °C at 9 V DC with 145 mA current [16]. Our result shows heating-paint paper heater can reach 60 °C in 0.5 second at 6 V DC with 315 mA current, which indicates about 22 times faster heating performance than silver-conductive ink heater despite lower voltage. Since the sustainable maximum temperature of polyurethan-based heating paint is 120 °C[9], we determined that a voltage range of 6 to 9 volts is a practical range to use for a small paper heater to melt a small amount of phase change materials utilized in the proposed stiffness switch, specifically PLA and hot melt adhesive.

3.3 H-beam Stiffness Switch

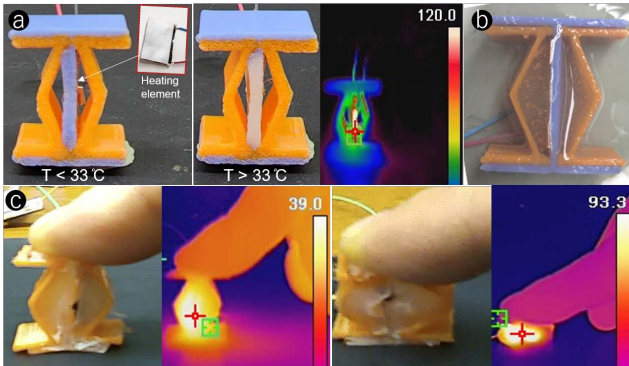


Figure 5: Prototyped H-beam stiffness switch.

As a proof of concept, we prototyped H-beam switch with dual-extruder FDM 3D printers (Raise 3D Pro3 plus and Snapmaker J1). With 100 % filament infilling condition, we fabricated the elastic frame with Polyflex TPU95 and H-beam with a TRONXY thermochromic PLA to visualize the heating process of the pillar (Figure 5 (a)). At room temperature the color of PLA pillar is blue and it becomes white as the temperature increases above 33 °C. The size of H-beam is 17 mm (W) × 19 mm (H) × 10 mm (D). The thickness of the elastic frame and H-beam is 1 mm and the maximum horizontal space between the pillar and side spring is 5 mm. We created a paper heater of 13 mm × 10 mm size and attached the coated side of the paper to the pillar with a double-sided heat transfer tape. After sealing the opposite side of the pillar with a molding oil clay, we infused silicon (Ecoflex 00-35 Fast) to the triangle side space of the pillar to provide shape recovery elasticity at soft mode from large deformation, as Figure 5 (b) shows. When we activated the heating element with 8 V of voltage and heated PLA pillar above the glass

transition temperature of PLA (i.e., 65 °C), the H-beam stiffness switch becomes soft enough to buckle by finger pressure after 30 seconds (Figure 5 (c)). We measured the elastic buckling strength of the prototyped H-beam switch with a digital force gauge (IMADA ZTS-DPU-500N) by compressing the structure with a CNC machine (Avalontech C-beam machine) as Figure 6 (a) and (b) shows.

We measured the buckling strength of the PLA-based H-beam in the temperature range from 27 °C to 90 °C with 10 °C steps as Figure 7 shows. At 0 input voltage, the H-beam shows a maximum buckling strength of 279.1 N. After applying an 8 V input voltage with 326 mA of current for 30 seconds, H-beam loses the buckling strength down to 21.4 N as Figure 6 (c) shows. The H-beam recovered the original shape immediately after the release of compression with the elasticity of the infilled silicon and the thermal shape memory effect of PLA, and regained the strong buckling strength (163 N) at room temperature (Figure 6 (b)).

Between 27 °C and 50 °C, the buckling strength of H-beam rapidly decreases from 279.1 N to 72 N. Above 50 °C, it gradually loses the buckling strength from 72 N to 21.4 N. This result indicates we can significantly change the effective stiffness of the H-beam stiffness switch at a lower temperature range than PLA glass transition temperature we utilized (i.e., 65 °C) and improves the efficiency of thermal stiffness control without requiring high voltage to gain large heating energy.

3.4 Linkage Joint Stiffness Switch

We fabricated a proof-of-concept of a linkage joint stiffness switch by 3D printing a regular PLA with FDM 3D printers (Raise 3D plus and Snapmaker J1) and utilized hot melt adhesive (BOSCH Ultra GS7ULT) with a melting temperature of 160 to 200 °C. We created two sets of 20 mm (L) × 4 mm (W) × 2 mm (D) size of linkage frames with a pillar and hole diameter size of 2 mm. We created a joint heating element by attaching copper foil tape and soldered wire on the side of the pillar as electrodes and painted the electrode and pillar with Carbo e-Therm heating paint. After drying the heating paint of the joint with a heat gun, we connected the frame joints together and infilled the joint space with hot melt adhesive using a BOSCH Cordless Glue Pen (Gluey). As Figure 8 (a) and (b) show, the linkage frames are fixed at room temperature and move together without rotating the pivot. Once the heating joint is activated by 8 V voltage, the hot melt at the link joint becomes liquid after 10 seconds and the pivot becomes flexible to rotate the end beam with a subtle proto-linkage torque force (10 N·cm) as Figure 8 (c) shows. We also fabricated a four-beam-linkage structure and found the deformation switch capability with the same principle (Figure 8 (d) and (e)).

4 FUTURE APPLICATION SCENARIO

We can integrate a stiffness switch into mechanical systems to regulate elasticity based on application context or facilitate a linkage mechanism control. In this section, we discuss four future application scenarios that demonstrate how the proposed stiffness switch can be used in product design and mechanical engineering.

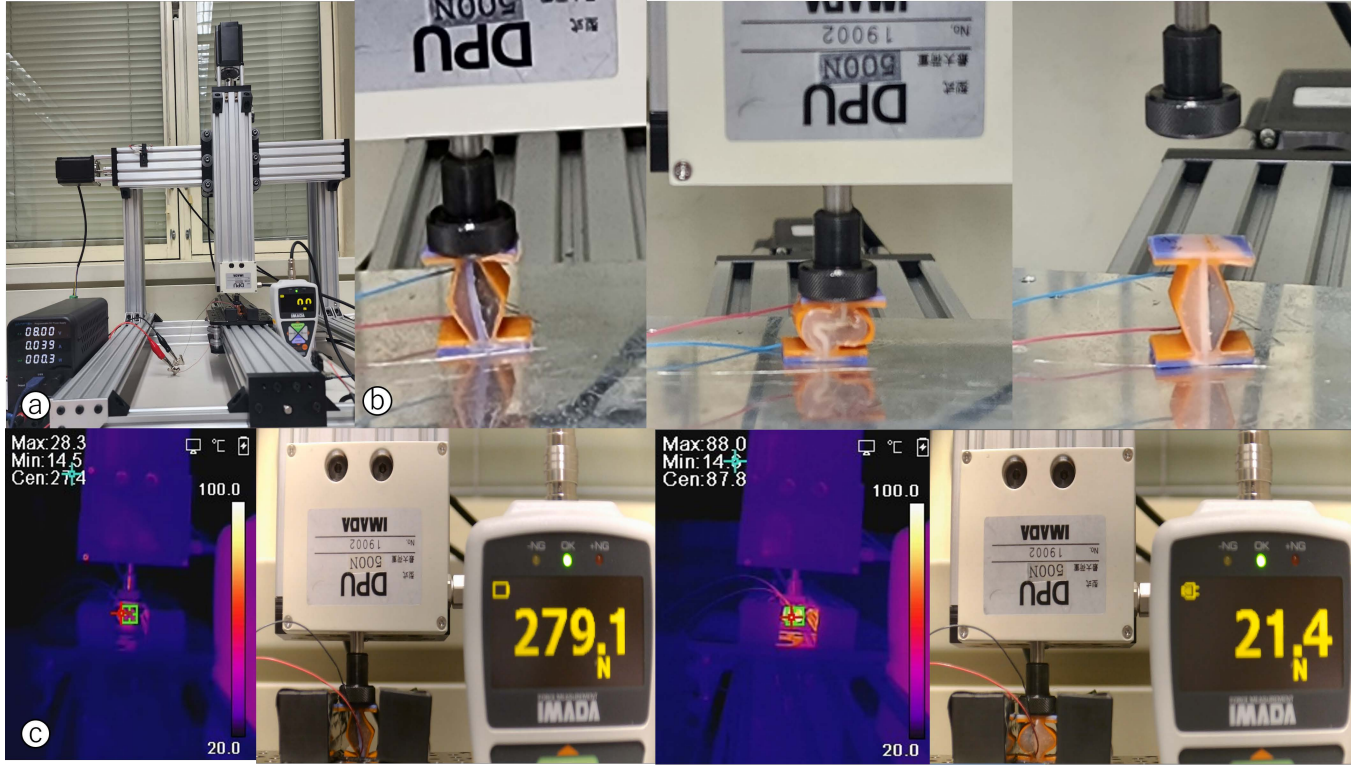


Figure 6: Compression test of H-beam stiffness switch.

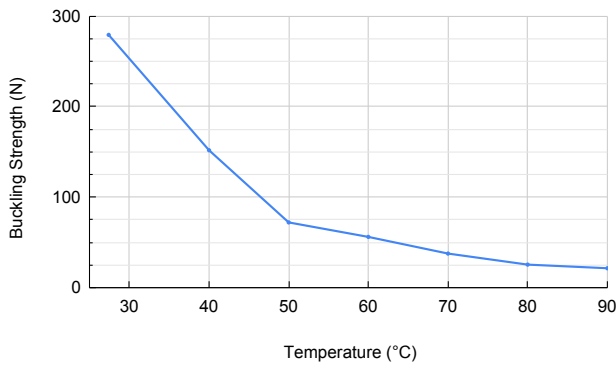


Figure 7: Buckling strength of PLA H-beam based on temperature.

4.1 Programmable Elastic Product

4.1.1 Adaptive Airless Bike Tire. An airless tire utilizing an elastic cellular structure has gained attention as a new product design solution to tackle the problem of a flat tire in a conventional pneumatic tire [13, 26]. While airless tires offer robust traction for off-road driving and cycling, as well as reduced weight, their fixed stiffness can make it challenging to balance both suspension capability and speed according to road conditions. We can utilize H-beam stiffness switches as programmable cells in airless tires, which can

switch the elasticity of the bike according to the cycling mode as Figure 9 (a) shows. For instance, to pedal along a flat, smooth road at a fast speed, we can turn off the H-beam stiffness switch and maximize stiffness to reduce friction between the tire and the road surface. On the other hand, when passing through a bumpy road, we can apply voltage to the H-beam stiffness switch and minimize the tire stiffness to achieve comfortable cycling with enhanced vibration-damping capability.

4.1.2 Tactile Tablet Pen. The H-beam stiffness switch can be used to control the elasticity of other materials by connecting them together in series. In soft mode, when the stiffness switch is softer than the combined material, it can absorb the compression stress and strain, making the stiffness of the other material imperceptible to the user. Conversely, in hard mode, when the stiffness switch gains significantly more stiffness than the other material, the user can feel the stiffness of the material with little compression loss. If we use two stiffness switch with two different elastic materials (e.g., foam, spring, elastic metamaterial unit cell [35]) in parallel, we can choose the elasticity from one of the combined materials. Based on this principle, we can create a tactile feedback tablet pen with elasticity that can be reconfigured as Figure 9 (b) shows. For instance, if we combine a sponge material and a hard spring with one stiffness switch each, we can choose the rigid feeling of the hard spring for ballpoint mode and the soft, elastic feeling of the sponge material for brush pen mode. The stiffness switch can also be used to select the elasticity from multiple elastic metamaterial unit

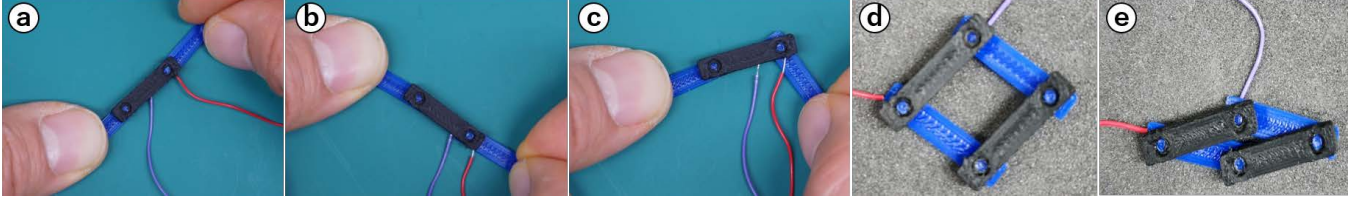


Figure 8: Prototype linkage joint stiffness switch and four-bar linkage.

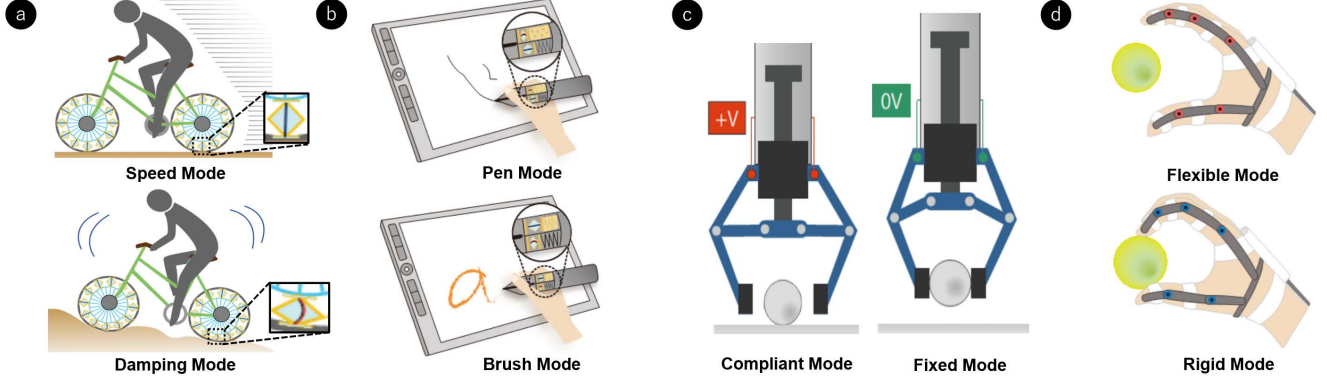


Figure 9: Application examples: (a) Adaptive airless bike tire deploying multiple H-beam stiffness switches that support high-speed cycling on flat roads and automatically switch to a damping state on bumpy roads for a smoother ride. (b) Tactile tablet pen developed with reconfigurable elasticity that allows users to switch between the feeling of a hard ballpoint pen and a soft brush by exchanging the stiffness of pre-defined elastic materials. (c) Robotic hand utilizing a linkage stiffness switch to control the softness and hardness of its grip. The mechanical link compliantly moves with a flexible joint to grasp an object according to the joint stiffness controlled by the input voltage, and then fixes the joint to hold the object securely by turning off the voltage. (d) Finger exoskeleton with multiple linkage stiffness switches that can provide haptic feedback of virtual objects and also augment the dexterity to grasp and hold fragile objects.

cells and enable dynamically reconfigurable elastic metamaterial devices [5, 17, 35].

4.2 Programmable Linkage Mechanism

4.2.1 Robotic Hand. Linkage mechanism has been utilized in many mechanical designs of robotic hands[18]. Conventional link joints are used as passive components of actuators to control the grip motion. In particular, a robot hand utilizing a linear actuator[1] requires a precise design of mechanical link with a complex control system to achieve the mechanical link to achieve specific grasping speed and holding torque. The proposed linkage joint stiffness switch can augment the link joint as active components and support the delicate grasping capability with electrical joint stiffness control as shown in Figure 9 (c). In a compliant mode, the link joint becomes flexible as the phase change material at the joint melts according to the applied voltage. By tuning the joint stiffness, we can convert the rapid motion of the linear actuator to a delicate grip motion to securely grab an object. Moreover, the robot hand can pick up objects without the need for power by maximizing joint stiffness through solidification of the phase change material, which can be achieved by stopping the voltage application to the linkage stiffness switch.

4.2.2 Finger Exoskeleton. Similar to the robot hand, the mechanical design of finger exoskeletons has been explored with linkage mechanisms in the field of human augmentation to enhance human dexterity and hand control capability with a compact and lightweight design[24]. The finger exoskeleton often requires extra frames and actuation space to utilize motors for finger joint control. We can improve the flexibility and wearability by directly controlling the stiffness of every linked joint of exoskeletons with phase change material, as shown in Figure 9 (d). This method supports the lightweight control of a series of linked joints to mimic the movement of natural fingers and provide a wide range of haptic feedback to manipulate virtual objects and facilitate delicate manipulation of real objects.

5 CONCLUSION

We proposed Stiff-switch, a low-voltage thermo-mechanical stiffness switch with shape recovery capability. The conventional method of thermal stiffness control requires high voltage to liquidity phase change materials and the stiffness control is not repeatable due to the lack of elasticity in phase change material. To resolve these issues, we introduced a novel thermo-mechanical structure that utilizes a minimum amount of phase change materials based on the

stress concentration for linear stress and torque. Specifically, we introduced an H-beam stiffness switch that uses phase change material to thermally control its buckling strength. To control torque stiffness, we presented the linkage joint stiffness switch, which deploys the phase change material to control the friction of the linkage pivot joint with temperature. As a low-voltage joule heating method, we discussed the fabrication method of a customizable carbon-based heater and demonstrated a high-speed heating performance for a small-scale to heat phase change materials efficiently. We fabricated the proof-of-concept with a 3D printer and silicone and demonstrated a wide range of stiffness control with shape recovery capability from large deformations. As the possible application scenarios, we discussed the deployment of multiple stiffness switches to dynamically reconfigure the elasticity of airless tires and tactile tablet pens and described the usability for robot hands with a programmable linkage mechanism.

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